

Write programs for each problems using for loop and while loop.

Q1) print all natural numbers from 1 to n.

input format: integer n
output format: 1
2
⋮
n

Q2) print all natural numbers from n to 1.

input format: integer n
output format: n
n-1
n-2
⋮
3
2
1

Q3) print all alphabets from a to z.

output format: a
b
c
⋮
z

Q4) print all alphabets from a to z and A to Z

output format: a
b
⋮
z
A
⋮
Z

Q5) print all even numbers between 1 to n.

n is a input

input format = integer n
Output " = 2
4
⋮
(n)

Q6) same as Q5, print all odd numbers.

Q7) find sum of all natural numbers from 1 to n

Q8) find (print) sum of all even numbers between 1 to n.

Q9) same as Q8, sum of all odd numbers.

Q10) print multiplication table of any input number

input format: integer n

for example:- n = 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
⋮
5 x 10 = 50

Q11) count the number of digits in a number.

Example: n = 125678

Output: 6

Q12) output the first and last digit of a number.

Example:- n = 125798

output:- The first digit is 1
The last digit is 8

Q13) swap (exchange) first digit and last digit of a number.

Example:- n = 123578

Output:- 823571

Q14) calculate sum of the digits of a number.

Example:- n = 123457

Output = 22 (1+2+3+4+5+7)

Q15) same as Q14). Calculate the product of digits.

Ex:- n = 1254

output: 60 (1x2x5x4)

Q16) implement a function to get the factorial value.

function Argument ~ Number
return ~ factorial value.

Q17) implement a function to generate the fibonacci sequence.

function Argument → Number
return → Sequence of numbers.

Ex:- fibonaci(3)

return → 0, 1, 1, 2

fibonaci(0) = 0

fibonaci(1) = 1

fibonaci(2) = 1

fibonaci(3) = 2

Fibonacci Sequence
0, 1, 1, 2, 3, 5, ...

Q18) make a function to check if a given thing is a palindrome or not?

function Argument → anything string or integer

return → true or false

reading it to the reverse doesn't make any change.

"MOM" "Civic" "rotor"

print these shapes

1001

Q19) input or function argument is no. of rows.

Ex: rows = 4

*
* *
* * *
* * * *

Q20) rows = 4

* * * *
* * *
* *
*

Q21) height = 5

width = ?

* * * * *
* * * * *
* * * * *

Q22) rhombus.

input or argument → side = 3

* * *
* *
*

Q23) rows = ?

Ex → 3

*
* *
* * *

Q24) Triangle.

input or argument = rows

Ex → rows = 5

*
* *
* * *
* * * *
* * * * *

Q25) rows = ? spaced triangle

Ex: rows → 5

*
* *
* * *
* * * *
* * * * *

Q26) rows → ?

rows → 5

*
\$ \$ \$
* * * *
\$ \$ \$ \$ \$
* * * * *

Q29) find the largest value in a given number